

Blockchain-Assisted Message Reporting Scheme With Weighted Threshold Signature for Vehicular Ad-Hoc Networks

Ru Li, Jie Cui , Senior Member, IEEE, Jing Zhang , Lu Wei , Member, IEEE, Hong Zhong , and Debiao He , Member, IEEE

Abstract—In vehicular ad-hoc networks (VANETs), message reporting is an effective method for improving traffic safety and efficiency. Most existing VANET message reporting schemes rely on the trust value of a single vehicle to determine message authenticity, which leads to unreliable message sources. Even multi vehicle-assisted reporting schemes are limited by the assumption that all vehicles have the same credibility, which does not reflect the actual dynamic VANET environment in which vehicles have different credibilities. To address this issue, we propose a blockchain-assisted VANET message reporting scheme with weighted threshold signatures. Through the design of weights, the credibility of different vehicles is quantified, and the impact of vehicles on the signing process is differentiated. Threshold signature generation relies on the weight sum of all signatories reaching a predetermined threshold, to enable flexible and reliable message reporting. Security analysis shows that our proposed scheme combined with blockchain can satisfy the security and privacy requirements of VANET message reporting. Performance analysis indicates that our proposed scheme outperforms the most advanced VANET message reporting schemes in terms of transmission and computation performance.

Index Terms—Blockchain, security authentication, VANETs, weighted threshold signature.

I. INTRODUCTION

WITH the development of information and intelligent technologies, vehicular ad-hoc network (VANET) technology has gradually become an important component of intelligent transportation. In VANETs, the onboard unit (OBU) achieves vehicle-to-vehicle (V2V) and vehicle-to-infrastructure

(V2I) information sharing through IEEE 802.11p or cellular networks. Unexpected events such as traffic accidents, road congestion, and temporary or adverse road conditions may frequently occur in practical applications. Therefore, message reporting is an essential part of VANET system for the timely reporting of emergencies to relevant departments, to ensure a quick response, that would reduce the risk of accidents, thereby improving road safety and traffic efficiency [1], [2], [3].

Communication in VANETs depends primarily on public channels. However, open wireless communication channels introduce security and privacy issues [4], [5], [6], such as attacker monitoring, interception, or tampering with normal messages in vehicle communication channels, which threatens road traffic safety or discloses driver privacy. Therefore, in vehicle message reports, the receiver must authenticate the message to ensure the security of the on-board communication channel. A well-designed message reporting scheme allows vehicles to first generate message signatures, which can be a single signature with its own reputation value or a threshold signature that requests assistance from multiple parties. The message and its signature are then reported to the relevant servers to directly ensure the reliability and accuracy of the message source.

Most existing message reporting schemes in VANETs are based on centralized single-signature authentication methods [7], [8], [9] that rely on the assumption that messages sent from vehicles with higher credibility are more trustworthy. Report receivers in such schemes perform only single-signature authentication, relying on tracking penalties for vehicles after false message reporting behavior occurs, and failing to identify the authenticity of messages in time, unless the receiver conducts a specific event aggregation analysis on all received messages. However, this approach consumes a large amount of storage resources at the receiving end, making it difficult to provide prompt feedback.

To address the issue of low credibility caused by centralized single-signature message authentication, some scholars have recently proposed collaborative multi-signature message reporting schemes. For example, Li et al. [10] proposed a blockchain-based vehicle cloud-assisted message reporting scheme that targets the security issues in announcement message dissemination by VANETs. The scheme utilizes threshold signature technology, in which vehicles can sign reported

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Ru Li, Jie Cui, Jing Zhang, Lu Wei, and Hong Zhong are with the Key Laboratory of Intelligent Computing and Signal Processing of Ministry of Education, School of Computer Science and Technology, Anhui University, Hefei 230039, China, and also with the Anhui Engineering Laboratory of IoT Security Technologies, Anhui University, Hefei 230039, China (e-mail: cuijie@mail.ustc.edu.cn).

Debiao He is with the School of Cyber Science and Engineering, Wuhan University, Wuhan 430072, China (e-mail: hedebiao@163.com).

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messages with the help of adjacent vehicles, generating trusted announcement messages. Concurrently, a reputation evaluation method for announcement messages is designed for trust management. Chen et al. [11] proposed a new message reporting architecture based on multiple signatures to address the issues of insufficient effectiveness and inability to provide secure reporting in existing message reporting schemes. This architecture improves message reporting efficiency by converting chameleon signatures into multiple signatures, thereby reducing the computational and communication costs of vehicle-side OBUs and RSUs.

Although multi-signature message reporting schemes for VANETs have significantly reduced the possibility of spreading false messages compared with centralized single-signature schemes, these schemes also have two main deficiencies. All vehicles perform multiple-signatures based on the principle of same credibility (i.e., equal weight values), resulting in a fixed signature permission control of the vehicle, which cannot meet the flexible and practical message reporting requirements in a dynamic VANET. In addition, issues associated with insufficient security or performance exist. Some schemes cannot effectively revoke identity [10] or provide identity privacy protection [11][12] [13], or cannot guarantee that a vehicle cannot be disguised as multiple legitimate vehicles. Moreover, some schemes fail to consider the computational load on the fog node side and cannot perform batch verification of the reported messages [14] [15]. In addition, the cost of message reporting signatures and verification in these schemes is relatively high.

A. Contributions

To address these limitations, we propose a blockchain-assisted weighted threshold signature scheme for message reporting in VANETs. By designing a three-factor reputation evaluation mechanism, different weights are assigned to vehicles, and based on different weight values, threshold signatures for messages are generated using elliptic curve digital signature algorithm (ECDSA). Through the flexible adjustment of the weight threshold, rapid message reporting after one round of signing is achieved, making this design especially suitable for scenarios with low vehicle density. In addition, blockchain is utilized to assist with identity authentication, privacy protection and revocation during the process of requesting auxiliary signatures and message reporting, thereby preventing various attacks and enhancing system security. The contributions of this study are as follows.

- 1) Considering that a threshold signature paradigm of the same weight cannot meet the requirements of dynamic VANET message reporting, a message reporting scheme based on weighted threshold signatures is proposed. Through the design of weights, different vehicles participating in the signature are processed differentially, and the weights of the vehicles are reduced or increased to control their impact on the signature process. By dynamically adjusting the threshold weight, the threshold signatures of messages can be generated rapidly to achieve more flexible and reliable message reporting.

- 2) To achieve security authentication during vehicle message reporting, a blockchain-assisted revocable authentication method is designed. Without the need for a trusted party, a vehicle can register its anonymous public key with the blockchain on demand. By utilizing the blockchain for identity registration, authentication, effective revocation of relevant entities, and managing the reputation value, our proposed scheme combines the blockchain with threshold signatures to enhance the security of the authentication process while protecting vehicle identity privacy.
- 3) Performance analysis results show that our proposed scheme constructed using elliptic curve cryptography combined with blockchain has better computational and transmission performance than other existing message reporting schemes for VANETs. The security analysis proves the security and privacy protection capability of our proposed scheme.

B. Organization of the Paper

The rest of the paper is structured as follows: Section II introduces related work, Section III provides background knowledge. In Section IV, details of the message reporting scheme for VANETs are described. Sections V and VI present detailed security analysis and performance analysis, respectively, and Section VII concludes the paper.

II. RELATED WORK

In VANETs, a series of research efforts have been conducted to achieve secure and efficient message reporting. These efforts can be mainly categorized into two types: centralized single-signature message reporting schemes and collaborative multi-signature message reporting schemes. As shown in Table I, we summarize and analyze the techniques, advantages, and limitations of several existing schemes.

A. Centralized Single-Signature Message Reporting Schemes

In the centralized single-signature message authentication scheme, it relies on the assumption that the message reporting vehicle itself is honest.

Wei et al. [16] proposed a privacy-preserving authentication scheme for reporting traffic emergency messages in VANETs. This scheme has lower communication overhead and requires less storage space. However, it is vulnerable to forged signatures. Limbasiya et al. [17] proposed a secure message communication scheme that uses low-cost primitives for message signing while maintaining user anonymity. Soleymani et al. [18] proposed a privacy-preserving message authentication scheme and trust model for VANETs, which addresses the issue of selfish nodes gaining benefits by disseminating false messages. However, the transmission overhead of this scheme is relatively high. Ahmed et al. [12] proposed a blockchain-based vehicle authentication and emergency message distribution and storage scheme, which reduces computational and storage consumption. However, the scheme does not consider vehicle privacy protection. Zhang et al. [19] proposed an efficient emergency

TABLE I
SUMMARY AND COMPARISON OF RELATED SCHEMES

	Scheme	Techniques	Advantages	Limitations
Centralized Architecture	[12]	* Pseudo random function * ECC-based * N-degree interpolated polynomial	* provide smaller storage space * provide lower communication cost	* exist the forged signature attacks
	[14]	* Bilinear pairing * Uotient filter	* prevent malicious nodes from spreading false message	* high transmission cost
	[15]	* Blockchain * Proof-of-Presence	* ensure emergency message authentication and transmission * reduce computing and storage cost	* no consideration for vehicle privacy protection
	[16]	* Space-Time Block Coding technology	* improve reliability of message broadcasting	* unnecessary communication cost * cause cache pressure
	[19]	* Blockchain * Reputation assessment model	* detect malicious behavior of vehicles * improve the reliability of message reporting	* unable to protect vehicle privacy
Collaborative Architecture	[20]	* Secret shared * Hash-chain * Blockchain	* implement distributed vehicle identity collaboration authentication	* unable to resist man-in-the-middle attacks * unable to resist replay attacks
	[21]	* Proof of work * Threshold signature	* avoid false message reports	* vehicle-side transmission cost increases linearly over time
	[10]	* Merkle Hash Tree * Threshold Signature * Smart contract	* provide reliable message transmission * implement vehicle privacy protection	* malicious vehicle identities cannot be effectively revoked
	[11]	* Multi-signature * Variant chameleon hash function * Chameleon signature	* support batch verification of message signatures	* unable to protect vehicle identity privacy * unable to resist replay attacks
	[22]	* Hash function * Bilinear pairing * Blockchain	* ensure the authenticity of messages * support conditional privacy protection for vehicle identities	* high communication and computing cost
	[24]	* ECC-based * Shamir secret shared * Threshold signature	* reduce the spread of false message * complete message signature generation and verification fastly	* no support for malicious vehicle tracing

alert message broadcasting scheme based on cooperative transmission, which considers the costs and benefits of cooperative transmission. However, this scheme introduces unnecessary communication overhead and caching pressure. Xu et al. [20] proposed a low-latency message authentication scheme based on message priority, which optimizes communication overhead and authentication efficiency through cooperation between RSUs and vehicles. Ahmed et al. [21] proposed a blockchain-based incentive trust model for VANETs, the possibility of providing false information is reduced through an incentive mechanism. Fernandes et al. [22] proposed a decentralized reputation system based on a consortium blockchain, which analyzes the reliability of vehicle alarm messages based on the vehicle reputation. However, this scheme cannot protect vehicle privacy.

B. Collaborative Multi-Signature Message Reporting Schemes

In the centralized single-signature message authentication scheme, the message receiver authenticates only the single source identity, and determines whether to receive the message or directly receive the message after passing the identity authentication. Although the message source credibility is ensured, the authenticity of the message itself cannot be recognized in time. To address this issue, some research proposals have put forward a collaborative multi-signature message authentication scheme, which enhances message credibility through multi-vehicle threshold signature reporting.

Liu et al. [23] proposed a distributed blockchain-enabled collaborative authentication scheme, where the transit station

entrusts multiple proxy vehicles to perform identity verification simultaneously with a driving vehicle. However, the scheme cannot resist man-in-the-middle attacks and replay attacks. Baza et al. [14] proposed a scheme that combines threshold signatures with proof-of-work to detect false messages. False reporting is avoided by using multiple consecutive RSUs as proof of the vehicle's anonymous location, but the transmission overhead on the vehicle side increases linearly over time. Li et al. [10] proposed a secure and trustworthy location-based message dissemination scheme, which utilizes threshold signature technology to generate trusted messages. However, malicious vehicle identity of this scheme cannot be effectively revoked. Chen et al. [11] proposed a novel OBU-friendly emergency reporting scheme based on multi-signatures, where the OBU only generates multi-signatures through simple operations and supports batch verification of signatures. However, this scheme cannot protect vehicle identity privacy or resist replay attacks. Zhang et al. [24] proposed a message reporting scheme combining certificate-free message authentication with blockchain incentives, introducing an adaptive threshold multi-signature mechanism to jointly verify message authenticity. However, the scheme has high computational and communication overhead. Gu et al. [25] proposed a message reporting scheme for VANETs to address efficiency and privacy issues, the reported message authenticity is guaranteed through multi-participant reporting. Yadav et al. [26] proposed a threshold signature message reporting scheme using the Schnorr signature scheme, only a small number of cryptographic operations are required to complete signature generation and verification. However, this scheme is unable to track malicious vehicles.

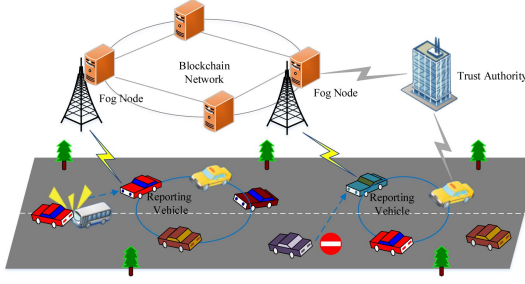


Fig. 1. The system model of our proposed scheme.

In summary, these state-of-the-art cooperative message authentication schemes still have deficiencies. Specifically, the vehicle entities in VANET are untrusted, and current message reporting schemes assume that all vehicles have the same credibility, which is not suitable for the real dynamic VANETs message reporting environment. In addition, these schemes incorporate the design of protecting vehicle identity privacy while ensuring message authenticity, which also poses security risks to multi-signature schemes. Therefore, it is necessary to design a flexible, secure, and efficient message reporting scheme for VANETs.

III. PRELIMINARIES AND BACKGROUND

A. System Model

We propose a framework for message reporting in VANETs, as shown in Fig. 1. This framework primarily consists of the following four entities.

- **TA:** Trusted Authorities (TAs) are regulated by government traffic management departments and distributed regionally. TAs are responsible for the registration, revocation and reputation management of all entities, and system initialization in the VANETs, and are considered to be absolutely trusted entity.
- **FN:** Fog nodes (FN) are deployed roadside based on demand and communicate with vehicles via wireless networks. They are responsible for receiving messages reported by vehicles, verifying the authenticity of vehicle identities and the reliability of messages, and preventing malicious users from reporting false messages.
- **Vehicle:** Each vehicle is equipped with an OBU, which is used for communication with other vehicles or fog nodes. Vehicles are considered untrusted entities in VANETs, so all messages sent by vehicles need to be verified by the receiver. In our scheme, vehicles are mainly divided into two types: auxiliary signature message vehicles and message reporting vehicles. Message reporting vehicles can also serve as auxiliary signature message vehicles.
- **Blockchain:** The blockchain system consists of fog nodes, which is responsible for storing the public information of the fog node and the vehicle (such as the communication public keys), and the announcement information further formed by the fog node after confirming the receipt of the message of the vehicle.

B. Preliminaries

Weighted threshold signature (WTS) technology can be used to generate authenticated signatures. The signature is constructed based on the Chinese Remainder Theorem (CRT) secret sharing technique and ECDSA algorithm [27]. Unlike traditional threshold signatures, which treat all participants equally and require a certain proportion of participants to be honest (such as those constructed based on Shamir's secret sharing [28]), WTS assign different weight values to participants, requiring only a certain proportion of the total weight of participants to be honest. The following describes the four methods involved in using the signature technology.

1) **Weighted Secret Sharing with Small Shares Based on CRT** $\{s_i\}_{i \in [n]} \leftarrow \text{Share}(\mathbf{P}, W_t, s)$: Let P be the base point of an elliptic curve group $\mathbb{G}$ of a prime order p_0 . Suppose we have a secret $s \in F_{p_0}$ and sets a public verification key as $pk = s \cdot P$. Each participant is associated with a weight value $w_i \in n$. We first select different integers p_i based on the weight values, where p_i are coprime numbers with binary length w_i , the value range is $2^{w_i} / (1 + 1/n) \leq p_i < 2^{w_i}$, and n represents the number of participants. Then, a random integer $S = s + u \cdot p_0$ is chosen, where u is uniformly distributed over the integer range $\{1, 2, \dots, I\}$, and $s_i = S \bmod p_i$ is calculated. Finally, the shares s_i of the secret s and the integer p_i are distributed to the participants $\mathbf{P} = \{P_1, P_2, \dots, P_n\}$. The secret s can be recovered only if the weight value of the participant exceeds W_t , where W_t is the weight threshold.

2) **Random Value Shares Generation** $[k]_i \leftarrow F_{\text{random}}(k_i)$: This method is primarily used when all participants jointly sample a random element. After the participants generate a signature for the message using their secret shares of the key, relevant random value shares, etc., it ensures that the aggregated signature that exceeds the weight threshold can be correctly verified by the verifier. Specifically, for $i \in [n]$, each participant P_i selects a random value $k_i \in F$ and computes the secret share of k_i using $\{[k_i]_j\}_{j \in [n]} \leftarrow \text{Share}(\mathbf{P}, T, t, k_i)$ and sends it to other participants. Each participant locally calculates the additive secret share $[k]_i = ([k_1]_i + [k_2]_i + \dots + [k_n]_i) \bmod p_i$ of the random value $k = k_1 + \dots + k_n \in F$ after receiving the share.

3) **Secret Recovery** $s \leftarrow F_{\text{recover}}(s_i)$: For a set of shares s_i that need to recover the secret value s , the Lagrange coefficient $\mathcal{L}_i = Z \cdot Z^{-1}$ is calculated by CRT theorem, where $Z = \prod_{j \neq i} p_j$ is the product of p_j except p_i , Z^{-1} is the multiplicative inverse of $Z \bmod p_i$, then $S = \sum_{i \in [x]} \mathcal{L}_i \cdot s_i \bmod P_x$ is calculated, where $P_x = p_1 \cdot p_2 \cdot \dots \cdot p_x$, x is the number of participants in recovering secret, and finally the secret s is recovered by $s = S \bmod p_0$.

4) **Secret Share Output** $[0 + out] \leftarrow F_{\text{open}}([out])$: As the secret sharing s_i of each participant is required to allow reconstruction of s , the secret sharing of 0 can be used to mask the secret sharing of out . (secret sharing simulation challenge in literature [27]). Specifically, each participant P_i holds a secret share $[out]$ of the output out and a secret share $[0]$ of 0 (obtained through the method F_{random}). Locally, each participant calculates and broadcasts a secret share $([0]_i + [out]_i) \bmod p_i$ of $0 + out$. After receiving the shares, each party reconstructs the value $0 + out$ as the value of out by using the method F_{recover} .

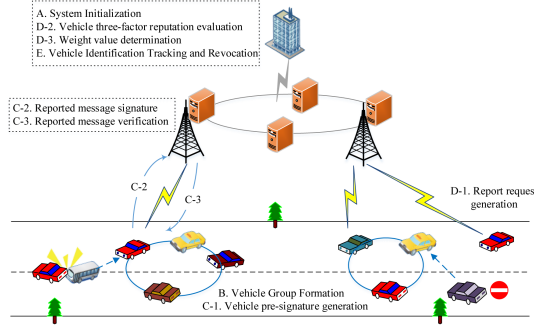


Fig. 2. Framework of our proposed scheme.

For all $i \in n$, the public input pk , and the message m , all participants can use their shares s_i to generate a signature on the message m . The signature generation process consists of a pre-signature phase and a signature phase. In the pre-signature phase, participants who are willing to sign interact with each other to generate and store relevant parameter values, thereby accelerating the signing process during real-time message reporting. In the signature phase, each participant uses the stored values to perform threshold signing on the reported message. Once the total weight of the aggregated signatures exceeds the predefined threshold, the verifier can successfully complete the verification.

IV. OUR PROPOSED SCHEME

In this section, we propose the framework of VANETs message reporting scheme as shown in Fig. 2. The first phase is initialization. TA initializes the system parameters and registers the vehicle and fog node. In the second phase, the vehicle group participating in the auxiliary signature message is formed in advance, so that the vehicle can report the message quickly. In the third phase, the vehicle reporting the message requests an auxiliary signature from the vehicles in the nearby group, the auxiliary vehicle uses the pre-signature to quickly generate the auxiliary signature, vehicles report messages and threshold signatures of the vehicle group to the fog node, and the fog node verifies the identity and signature of the messages reported by the vehicles. If the verification is successful, the fog node uploads the messages to the blockchain for sharing with other fog nodes. The fourth phase is to design a reputation evaluation mechanism, in which the weight value is determined based on the vehicle's reputation value. The fifth phase involves the tracking and revocation of malicious vehicles, the blockchain address of the malicious vehicle as well as the registered communication public key are completely deleted using the blockchain. Table II lists all symbols and their explanations used in our proposed scheme.

A. System Initialization

At this stage, TA first sets the system parameters, and then distributes relevant secret materials and parameters to fog nodes and vehicles during the registration process.

1) System Setup:

TABLE II
NOTATIONS

Notations	Definitions
α, P_{pub}	System private key and public key
W_n, W_t	The total weight value and the threshold weight of n vehicles
V_i	i -th vehicle
VID_i	V_i 's real identity
$Addr_i$	V_i 's blockchain address
vs_{ki}, VPK_i	V_i 's communication private key, and public key
Fog_j	j -th fog node
fsk_j, FPK_j	Fog_j 's communication private key, and public key
m_i	The reporting message sent by V_i
$H_i, (i = 0, 1, 2)$	Three secure one-way hash function

- TA generates an elliptic curve $E : y^2 = x^3 + ax + b \mod p$ with a security parameter λ , randomly selects coefficients $a, b \in F_p$, and then selects a base point P to generate a q -order elliptic curve group G ;
 - TA selects a random number α as the system private key, calculates the corresponding public key $P_{pub} = \alpha \cdot P$, encodes the public key into an address $Addr_{ta}$ using the Keccak-256 algorithm, and then registers it to the blockchain by invoking a smart contract. All TA addresses cannot be modified after registration;
 - TA selects several one-way hash functions: $H_0 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$, $H_1 : \{0, 1\}^* \times G \times G \times \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$, $H_2 : \{0, 1\}^* \times G \times G \times \mathbb{Z}_q^* \times G \times \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$;
 - TA stores α secretly, publishes system public parameters $param : \{p, q, a, b, P, P_{pub}, H_0, H_1, H_2\}$.
- 2) *Entity Registration:* During this phase, fog nodes and vehicles register with the TA through secure and private channels. Fog nodes can register through an encrypted communication channel using TLS/SSL, while vehicles register with the TA before leaving the factory [29], assuming that the TA has already completed its registration in the blockchain.

Fog Node Registration: Fog_j selects a unique real identity FID_j and submits a registration request $\{FID_j\}$ to TA. Upon receiving the request, TA will check whether the identity has been registered before. If not, TA selects a random number $sk_j \in \mathbb{Z}_q^*$, calculates $PK_j = sk_j \cdot P$, and encodes the public key into an address $Addr_j$. Then, TA uses its private key α and the ECDSA algorithm to generate a signed transaction $tx_1 = \{Addr_j\}_\alpha$ that is sent to the blockchain. Fog_j can select a random number fsk_j , calculate $FPK_j = fsk_j \cdot P$, register the public key FPK_j and expiry date EXP_j to the blockchain using the address $Addr_j$.

Vehicle Registration: Vehicle V_i selects the random key to calculate the corresponding address $Addr_i$, and submits the registration request $\{VID_i, Addr_i\}$ to TA together with the real identity VID_i (including driver's license number, license plate number, and other relevant information). TA checks whether the identity has been registered after receiving the request. If not, TA generates an initial reputation value R_i^{ini} , stores $(VID_i, Addr_i)$ locally, and sends the transaction $tx_2 = \{R_i^{ini}, Addr_i\}_\alpha$ to the blockchain. Vehicle V_i can choose a random number vs_{ki} , calculate the communication public key $VPK_i = vs_{ki} \cdot P$, repeat the above operations to obtain multiple public keys, and utilize the address $Addr_i$ to send public

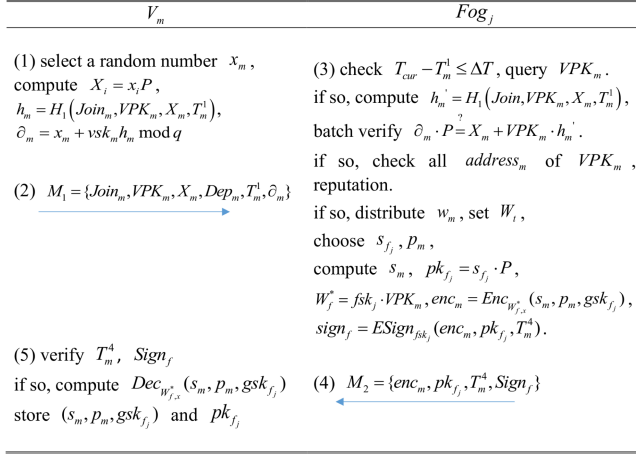


Fig. 3. Workflow of vehicle group formation phase.

keys $\{VPK_i^1, VPK_i^2, \dots, VPK_i^M\}$ to the blockchain for registration.

B. Vehicle Group Formation

As message reporting in VANETs requires as low latency as possible, it is necessary to consider completing some calculations in advance to reduce the real-time computational cost during the reporting process. Therefore, it is essential to establish a vehicle group to distribute the relevant keys and parameters for threshold signature. As shown in Fig. 3, the main steps of this phase are illustrated.

- 1) When vehicle V_m enters the communication range of a fog node Fog_j , all vehicles in the range can choose whether to join the vehicle group to assist in signing the message according to their wishes, so as to get rewards from the system after the message is successfully authenticated. If the vehicle wants to participate in the auxiliary signature message, a random number $x_m \in \mathbb{Z}_q^*$ is selected first, and public key $X_m = x_m P$, hash value $h_m = H_1(Join_m, VPK_m, X_m, T_m^1)$ and signature $\partial_m = x_m + \nu sk_m h_m \bmod q$ are calculated, where $Join$ and T_m^1 represent the request and timestamp of joining the vehicle group respectively. Then, the vehicle sends the join request $M_1 = \{Join_m, VPK_m, X_m, Dep_m, T_m^1, \partial_m\}$ along with a deposit Dep_m to Fog_j .
- 2) After receiving request messages M_1 from multiple vehicles within a time slot, Fog_j establishes a vehicle group. Firstly, the timestamp freshness is judged through $T_{cur} - T_m^1 \leq \Delta T$, where T_{cur} is the current timestamp, ΔT represents the max valid transmission interval. Then, it queries the blockchain to check if the public key VPK_m exists (for example, using smart contract technology [30]). If it exists, $h'_m = H_1(Join_m, VPK_m, X_m, T_m^1)$ is calculated separately, and the batch verification is conducted to see if the equality $\partial_m \cdot P \stackrel{?}{=} X_m + VPK_m \cdot h'_m$ holds. If it does, it checks if there are any duplicates in the registered addresses $Addr_m$ of all the public keys (which can be

obtained from logs) to determine if all the requests (via communication public keys) are from different vehicles. If the verification is successful, $Addr_m$ is used to query the reputation values of all vehicles within the vehicle group in the blockchain. Since the size of the final threshold signature is related to the weight values of all participants, to ensure security while avoiding excessive communication cost, the total weight W_n of the vehicle group will be controlled within a specific range. Based on the reputation value, Fog_j assign a weight value w_m and set the threshold weight as W_t (e.g., $W_t = W_n/2$). Then, Fog_j select a shared secret s_{f_j} and an integer p_m (where the binary length of p_m is w_m), and calculate the verification key as $pk_{f_j} = s_{f_j} \cdot P$. Using WTS technology to compute the corresponding secret shares $s_m, m \in [1, n]$, where n is the number of vehicles in the vehicle group. Finally, Fog_j calculate $W'_f = fsk_j \cdot VPK_m$, encrypt the secret shares s_m , integer p_m , and session key $gsk_{f_j} \in \mathbb{Z}_q^*$ within the vehicle group to obtain $enc_m = Enc_{W'_{f,x}}(s_m, p_m, gsk_{f_j})$, calculate $sign_f = ESign_{fsk_j}(enc_m, pk_{f_j}, T_m^4)$, where $ESign_{fsk_j}(\cdot)$ denotes the ECDSA signature algorithm, and return $M_2 = \{enc_m, pk_{f_j}, T_m^4, sign_f\}$ to the vehicles participating in assisting in signing the message within the vehicle group. It is worth noting that based on the reputation value or traffic conditions, some vehicles with low reputation values can be dynamically excluded from Fog_j , i.e., no weight is assigned to them.

- 3) After the vehicle V_m in the group receives the message M_2 from Fog_j , it first checks the timestamp freshness, and then uses the ECDSA verification algorithm and the public key FPK_j to verify the validity of the signature $sign_f$. If the verification is successful, it proceeds to decrypt enc_m by computing $Dec_{W'_{f,x}}(s_m, p_m, gsk_{f_j})$, and stores the decrypted (s_m, p_m, gsk_{f_j}) and pk_{f_j} locally.

C. Vehicle Message Reporting and Verification

The vehicle message reporting and verification phase consists of three parts: vehicle pre-signature generation, reported message signature, and reported message verification. As shown in Figs. 4 and 5, the main steps of this phase are illustrated.

- 1) *Vehicle Pre-signature Generation*: To reduce the latency in the process of real-time vehicle reporting, our proposed scheme will complete part of the signature operation in advance in the vehicle. Vehicle V_i first generates two random values $[r]_i \leftarrow F_{random}(r_i)$ and $[l]_i \leftarrow F_{random}(l_i)$, and broadcasts $l_i \times P$. Specifically, it uses the shares $\{[r]_i\}_{i \in [n]}, \{[l]_i\}_{i \in [n]}$ of r_i, l_i to calculate $enc_{v_i} = Enc_{gsk_{f_j}}([r]_i, [l]_i)$ and $sign_v = ESign_{vsk_i}(enc_{v_i}, VPK_i, T_i^1)$, resulting in $se_i = \{enc_{v_i}, VPK_i, T_i^1, sign_v\}$. Then, it sends se_i to other vehicles within the group. After receiving the message se_i , other vehicles verify the signature $sign_v$ and decrypt enc_{v_i} , and each vehicle calculates to obtain $[r]_i$ and $[l]_i$. Next, they calculate $[\theta]_i = ([r]_i * [l]_i) \bmod p_i$ and $[\delta]_i = ([r]_i * [s]_i) \bmod p_i$, encrypt and sign $[\theta]_i$, and share it with other participating vehicles. Upon receiving $[\theta]_i$, other vehicles perform local calculations to obtain $\theta = F_{open}([\theta]_i)$. Finally, all vehicles calculate $L =$

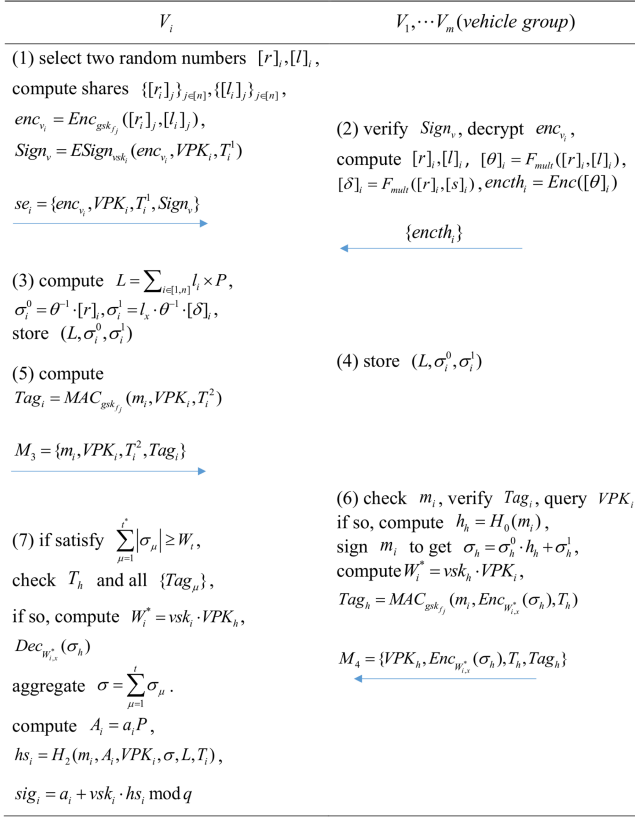


Fig. 4. Workflow of vehicle message reporting phase.

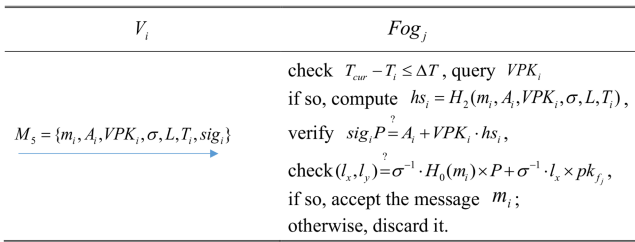


Fig. 5. Workflow of vehicle message verification phase.

$\sum_{i \in [1, n]} l_i \times P$ and set the elliptic curve point $L = (l_x, l_y)$. Each vehicle calculates $\sigma_i^0 = \theta^{-1} \cdot [r]_i$, $\sigma_i^1 = l_i \cdot \theta^{-1} \cdot [\delta]_i$ and saves the value $(L, \sigma_i^0, \sigma_i^1)$ for subsequent rapid signing of reported messages.

2) *Reported Message Signature*: When vehicle V_i encounters an emergency event during driving and needs to report message m_i to the fog node Fog_j within its range, $Tag_i = MAC_{gsk_{f_j}}(m_i, VPK_i, T_i^2)$ is first calculated and message $M_3 = \{m_i, VPK_i, T_i^2, Tag_i\}$ is broadcast to nearby vehicles to request auxiliary verification. Nearby vehicles (such as V_h) voluntarily choose whether to assist signature after receiving the request message M_3 . If V_h agrees to sign, it will check the legitimacy of message m_i , verifies Tag_i with the session key gsk_{f_j} and queries the validity of VPK_i in the blockchain after the check is passed. If the tag verification and public key query are correct, V_h will calculate the hash $h_h = H_0(m_i)$, sign the

message m_i with the previously saved value σ_h^0, σ_h^1 to create $\sigma_h = \sigma_h^0 \cdot h_h + \sigma_h^1$, and compute the value $W_i^* = vsk_h \cdot VPK_i$ and tag $Tag_h = MAC_{gsk_{f_j}}(m_i, Enc_{W_i^*}(\sigma_h), T_h, VPK_h)$. Then, V_h will send the assisted signature response $M_4 = \{VPK_h, Enc_{W_i^*}(\sigma_h), T_h, Tag_h\}$ to vehicle V_i .

Vehicle V_i receives auxiliary signature responses M_4 until the total weight value of signatures exceeds the threshold W_t (i.e., $\sum_{\mu=1}^t |\sigma_\mu| \geq W_t$, the sum of the sizes of the μ signatures σ_μ exceeds W_t), where t represents the total number of signatures. Firstly, the timestamp freshness and the validity of all tags $\{Tag_\mu\}$ are checked. If the check passes, $W_i^* = vsk_i \cdot VPK_h$ is calculated, $Dec_{W_i^*}(\sigma_h)$ is used to decrypt all collected signatures σ_μ , and all signatures are aggregated into $\sigma = F_{open}(\sigma_\mu)$, forming an aggregated signature (σ, L) for the message m_i . Then, V_i select an unused private key vsk_i and public key VPK_i , select random number $a_i \in \mathbb{Z}_q$, calculate $A_i = a_i P$, hash value $hs_i = H_2(m_i, A_i, VPK_i, \sigma, L, T_i)$, and signature $sig_i = a_i + vsk_i \cdot hs_i \bmod q$. Finally, the message, related public key, and signature are packaged into $M_5 = \{m_i, A_i, VPK_i, \sigma, L, T_i, sig_i\}$ and reported to Fog_j .

3) *Reported Message Verification*: After receiving the reported message M_5 , Fog_j first utilizes the equation $T_{cur} - T_i \leq \Delta T$ to determine the freshness of the timestamp T_i . If the equation does not hold, the message will be discarded. Otherwise, Fog_j checks if VPK_i exists in the blockchain. If VPK_i exists, Fog_j computes $hs_i = H_2(m_i, A_i, VPK_i, \sigma, L, T_i)$ to verify the equation $sig_i P \stackrel{?}{=} A_i + VPK_i \cdot hs_i$. Then, Fog_j calculates the equation $LP = \sigma^{-1} \cdot H_0(m_i) \times P + \sigma^{-1} \cdot l_x \times pk_{f_j}$, and determines whether $LP_x \stackrel{?}{=} l_x$ is equal to verifies the validity of the aggregate signature. If the total weight $W_n = w_1 + w_2 + \dots + w_t$ of the aggregate signature exceeds the threshold W_t (i.e., $W_n \geq W_t$), the verification will pass. Otherwise, Fog_j will directly discard the message. Once the reported message is verified and received, the reporting vehicle and the assisting signature vehicle will receive system rewards (identified through their current communication public keys). Meanwhile, Fog_j uploads the message to the blockchain to share with other fog nodes across the network, and these fog nodes will further broadcast it to other vehicles to improve traffic safety and efficiency.

Batch verification: Our proposed scheme supports batch verification, which means that Fog_j can simultaneously verify multiple messages reported from vehicles. After receiving multiple messages $M_5 = \{m_i, A_i, VPK_i, \sigma, L, T_i, sig_i\}$, Fog_j first check the timestamps freshness in the messages and conduct a batch query of the validity of the public keys VPK_i . Then, Fog_j employ the small-exponent testing technique [31] for batch signature verification. That is, a vector $v = \{v_1, \dots, v_i, \dots, v_u\}$ is randomly selected, where $v_i \in [1, 2^\nu]$, ν is a small random integer. Then calculate $hs_i = H_2(m_i, A_i, VPK_i, \sigma, L, T_i)$ and $h_i = H_0(m_i)$. Finally, batch verification is performed through formulas (1) and (2).

$$\sum_{i=1}^u v_i sig_i \cdot P = \sum_{i=1}^u v_i \cdot A_i + \sum_{i=1}^u v_i hs_i \cdot VPK_i \quad (1)$$

$$\sum_{i=1}^u L_i = \sum_{i=1}^u (\sigma_i^{-1}) h_i \cdot P + \sum_{i=1}^u (\sigma_i^{-1}) l_i \cdot pk_{f_j} \quad (2)$$

D. Weight Determination of Vehicle Threshold Signature

The level of vehicle's reputation directly affects the allocation of threshold signature weight when assisting in signing reported messages. Inspired by works such as [32] and [33], a three-factor reputation evaluation method is designed, and then an integer linear programming algorithm is utilized to determine the weight value based on the reputation value. The process of determining the reputation value and weight value is as follows.

1) *Report Request Generation*: TA first calculates the reputation of each vehicle, which is mainly determined by three factors: historical reputation, behavioral feedback, and group historical reputation. Historical reputation represents the previous reputation value of the reported vehicle, behavioral feedback means that vehicles within the system can collect evidence to report other vehicles with inappropriate behavior, and group historical reputation represents the reputation value of vehicles within the reporting group. Let R_i^{ini} be the initial reputation of vehicle i to address the cold start issue. Other vehicles in the system can form reporting groups to collect evidence and report on the bad behavior of vehicles, generating the following report message to submit to the TA, as formula (3).

$$rep_r = \{VPK_{i^*}, \{VPK_i\}_{i=1}^{N_r}, \zeta_r, Evid, Fee, T_r\} \quad (3)$$

where, VPK_{i^*} is the public key of the reported node i^* . $\{VPK_i\}_{i=1}^{N_r}$ are the public keys and ECDSA signatures of multiple reporting vehicles, $Evid$ is the evidence collected, such as requests, signatures, etc., Fee is the fixed reporting fee to prevent repeated reporting, and T_r is the timestamp.

2) *Vehicle Three-factor Reputation Evaluation*: After receiving the report message rep_r , TA will verify the signature and relevant evidence of the reported vehicle. Upon successful verification, it will deduct a penalty of Δ_{dec} from the reputation of the reported vehicle, and increase the reputation of the reporting vehicle by Δ_{inc} as well as a monetary reward. Generally, the increase of benign behavior will improve the reputation value of the vehicle, and the increase of malicious behavior will reduce the reputation value. Current behavior has a greater impact on reputation than historical behavior. Let Δ_b represent the reputation reward or penalty for behavior b , and define a binary variable $\lambda_b = \{-1, 1\}$, where $\lambda_b = 1$ if b is a benign behavior, and $\lambda_b = -1$ if b is a malicious behavior. In each time slot, the reputation of vehicle i is calculated using formula (4).

$$R_i^k = \sum_{b=1}^{N_i^k} (\lambda_b e^{-\varphi R_j^{k-1}} \Delta_b) + e^{-\gamma R_i^{k-1}} \quad (4)$$

where, N_i^k is the number of behaviors that node i is fed back in the k time slot, that is, the number of reported vehicles. γ and φ are non-negative exponential attenuation factors. R_i^{k-1} is the historical reputation of reported vehicles, R_j^{k-1} is the historical reputation value of vehicles in the reporting group.

After each time slot reporting behavior is established, TA sends $\{R_i^k, Addr_i\}$ to the blockchain to update the vehicle's reputation value and provide a basis for the fog node to assign weight values.

TA will divide three different reputation value intervals according to the reputation value of all vehicles in advance, such as $[-5, 0]$, $[0, 5]$ and $[5, 10]$, each interval represents different weight. When selecting the index decay factor γ , TA will make the selection based on the historical reputation value interval. The higher the reputation value (approaching the reputation boundary set by TA), the higher the γ . Conversely, the lower the reputation value, the lower the γ , so that the calculation of the current reputation value achieves the purpose of "rapid decline and slow growth". The selection of φ is inversely proportional to the reputation value of the vehicles in the reported group.

3) *Weight Value Determination*: TA will also divide the weight value into three intervals in advance, and the interval division of the reputation value and the weight value will be dynamically adjusted as needed. After the fog node queries the reputation values of the vehicles in the auxiliary signature vehicle group from the blockchain, it sorts them in descending order. Weight values are allocated based on reputation values. Since the size of weight values affects the communication cost of our scheme, the total weight value W_n needs to be controlled. Moreover, there may be significant differences in reputation among vehicles that need to be allocated weights, so it is necessary to ensure that each vehicle's weight falls within the specified range. Therefore, this problem can be expressed as an integer linear programming problem. The objective function for weight allocation is shown in formula (5), where n represents the number of vehicles allocated with weights, and b represents the fixed reference threshold signature size (bits/bytes) for a single vehicle. The constraint conditions are shown in formula (6), where $range_l$ and $range_h$ respectively represent the upper and lower bound values of the weight interval in which they are located, and $w_i \neq w_j$ represents that assuming that all vehicle reputation values are different, the distribution of weight values is also unique.

$$\lim \sum_{i=1}^n w_i = n * b \quad (5)$$

$$\begin{cases} w_i \in [range_l, range_h] \\ w_i \neq w_j, i \neq j \end{cases} \quad (6)$$

Finally, a set of weights that meet the conditions is obtained by solving the problem with the existing Coin-or Branch and Cut Solver (CBC) [34]. Thus, the weights of vehicles in the threshold signature are determined.

E. Vehicle Identification Tracking and Revocation

When a malicious vehicle reports false messages or assists in signing false messages, TA can obtain the blockchain address $Addr_i$ based on the vehicle's public key VPK_i , further tracking its real identity VID_i and imposing corresponding punishment. In the event that the reputation score drops below the set minimum threshold or direct identity revocation is required, the

generated public key needs to be revoked. Our proposed scheme first revokes the corresponding address $Addr_i$ to stop the further registration of the communication public key VPK_i through the blockchain-assisted method. Subsequently, TA packages all the communication public keys registered at the address $Addr_i$ into $vpklist = \{VPK_i^1, VPK_i^2, \dots, VPK_i^M\}$ and reputation information, and sends transaction $tx = \{vpklist, R_i^k\}_\alpha$ to the blockchain to request deletion. Finally, the relevant public keys and reputation values in the blockchain and blockchain address are deleted to achieve complete revocation.

V. SECURITY PROOF AND ANALYSIS

In this section, we first demonstrate the existential unforgeability under chosen-message attack (EU-CMA) security of our scheme through security proofs. Then, we prove through informal security analysis that our message reporting scheme for VANETs is capable of resisting various attacks.

A. Security Proof

Theorem 1: If the elliptic curve discrete logarithm problem (ECDLP) is (τ, ε) -hard in the elliptic curve group $\mathbb{G}$, then our scheme is (τ', ε') -hard under the random oracle and EU-CMA models, where the relationships between ε and ε' , τ and τ' are shown in formulas (7) and (8), respectively.

$$\varepsilon \geq \varepsilon_1 \cdot \left(\frac{\varepsilon_1}{q_{H_0} + q_{H_2} + q_E} - \frac{1}{q} \right) \quad (7)$$

$$\tau \approx 2\tau' + 2t_h + (q_S + 1)(4t_{ecm} + 2t_{epa}) \quad (8)$$

where, $\varepsilon_1 = \varepsilon' - \frac{(q_{H_0} + q_{H_2} + q_E)q_S}{q}$. q_{H_i} ($i = 0, 2$) represents the maximum number of hash queries, q_E represents the maximum number of queries for extracting the verification key, q_S represents the maximum number of signature queries, t_h represents the execution time for the hash function operation, and t_{ecm} and t_{epa} represent the execution times for scalar multiplication and point addition operations on the group $\mathbb{G}$, respectively.

Proof: Assuming there exists a polynomial-time adversary $\mathcal{A}$ who can break our proposed scheme with a non-negligible advantage ε' , that is, forge a valid message $\{m_i, A_i, VPK_i, \sigma, L, T_i, sig_i\}$, then the challenger $\mathcal{C}$ would be able to solve the ECDLP with a non-negligible advantage ε . Let $(P, Q = x \cdot P)$ be an instance of the discrete logarithm problem, and the goal of the challenger $\mathcal{C}$ is to compute x given P and Q . The adversary $\mathcal{A}$ can make oracle queries to the challenger $\mathcal{C}$, as described below.

Setup-Oracle: Upon receiving the query, Challenger $\mathcal{C}$ sets up $param$ and returns the public parameter $param : \{p, q, a, b, P, P_{pub}, H_0, H_1, H_2\}$ to Adversary $\mathcal{A}$.

Hash₀-Oracle: The challenger $\mathcal{C}$ predefines a set L_{H_0} in the form of $\langle m_i, l_0 \rangle$ and initializes the set L_{H_0} to be empty. Upon receiving an $Hash_0$ query initiated by the adversary $\mathcal{A}$ using $\{m_i\}$, the challenger $\mathcal{C}$ checks whether $\{m_i\}$ exists in the set L_{H_0} . If it exists, Challenger $\mathcal{C}$ queries the corresponding l_0 and returns it to adversary $\mathcal{A}$; Otherwise, $\mathcal{C}$ selects a random value $l_0 \in \mathbb{Z}_q^*$ (where, $l_0 \notin L_{H_0}$), $l_0 = H_0(m_i)$, inserts $\langle m_i, l_0 \rangle$ into the set L_{H_0} , and returns l_0 to adversary $\mathcal{A}$.

Hash₂-Oracle: The challenger $\mathcal{C}$ predefines a set L_{H_2} in the form of $\langle m_i, A_i, VPK_i, \sigma, L, T_i, l_2 \rangle$ and initializes the set L_{H_2} to be empty. Upon receiving an $Hash_2$ query initiated by the adversary $\mathcal{A}$ using $\{m_i, A_i, VPK_i, \sigma, L, T_i\}$, the challenger $\mathcal{C}$ checks whether $\{m_i, A_i, VPK_i, \sigma, L, T_i\}$ exists in the set L_{H_2} . If it exists, Challenger $\mathcal{C}$ queries the corresponding l_2 and returns it to adversary $\mathcal{A}$; Otherwise, $\mathcal{C}$ selects a random value $l_2 \in \mathbb{Z}_q^*$ (where, $l_2 \notin L_{H_2}$), $l_2 = H_2(m_i, A_i, VPK_i, \sigma, L, T_i)$, inserts $\langle m_i, A_i, VPK_i, \sigma, L, T_i, l_2 \rangle$ into the set L_{H_2} , and returns l_2 to adversary $\mathcal{A}$.

Extract Verification Key-Oracle: The challenger $\mathcal{C}$ predefines a set L_{vk} in the form of $\langle VPK_i, l_{v1} \rangle$ and initializes the set L_{vk} to be empty. Upon receiving an verification key query initiated by the adversary $\mathcal{A}$ using $\{VPK_i\}$, the challenger $\mathcal{C}$ checks whether $\{VPK_i\}$ exists in the set L_{vk} . If it exists, Challenger $\mathcal{C}$ queries the corresponding l_{v1} and returns it to adversary $\mathcal{A}$; Otherwise, $\mathcal{C}$ selects a random value $l_{v1} \in \{0, 1\}$ (where, $l_{v1} \notin L_{vk}$), inserts $\langle VPK_i, l_{v1} \rangle$ into the set L_{vk} , and returns l_{v1} to adversary $\mathcal{A}$.

Sign-Oracle: The challenger $\mathcal{C}$ presets a set L_{sig} of the form $\langle A_i, \sigma, L \rangle$ and initializes the set L_{sig} as empty. Upon receiving a signature query initiated by an adversary $\mathcal{A}$ using $\{m_i, VPK_i, T_i\}$ (assuming $\mathcal{A}$ has already performed the extract verification key query). The challenger $\mathcal{C}$ first selects random numbers $\sigma \in \mathbb{Z}_q^*$, $sig_i \in \mathbb{Z}_q^*$, $L \in \mathbb{G}$, and computes $l_2 = H_2(m_i, A_i, VPK_i, \sigma, L, T_i)$, $A_i = -sig_i P + VPK_i \cdot l_2$. If $H_0(m_i)$ or $H_2(m_i, A_i, VPK_i, \sigma, L, T_i)$ already exist, then $\mathcal{C}$ terminates the interaction, and sets a flag $bad \leftarrow true$. Finally, the challenger $\mathcal{C}$ checks if $\langle A_i, \sigma, L \rangle$ exists in L_{sig} , and if not, inserts $\langle A_i, \sigma, L \rangle$ into the set L_{sig} and returns $\langle m_i, A_i, VPK_i, \sigma, L, T_i \rangle$ to the adversary $\mathcal{A}$. After completing the above query process, the challenger $\mathcal{C}$ can generate a valid signature sig_i . The signature generated in the simulation is indistinguishable from the signature generated by the real protocol from the perspective of the adversary.

Forgery: After the above queries are completed, the adversary $\mathcal{A}$ outputs a forged message and signature $\{m_i, A_i, VPK_i, \sigma, L, T_i, sig_i\}$ to the challenger $\mathcal{C}$. Upon receiving the forged message and signature, the challenger $\mathcal{C}$ executes the authentication algorithm described in Section IV-C to verify the validity of the forged signature. If it is invalid, the interaction is terminated; otherwise, the challenger $\mathcal{C}$ considers that $\mathcal{A}$ has successfully forged the message $\{m_i, A_i, VPK_i, \sigma, L, T_i, sig_i\}$.

Let E be the event that the adversary $\mathcal{A}$ successfully forges a message. The occurrence of event E requires two conditions to be met: the forgery by the adversary $\mathcal{A}$ is legitimate and valid (denoted as E_f), and the challenger $\mathcal{C}$ does not terminate during the challenge phase (denoted as E_a). Then, the probability of $\mathcal{A}$ successfully forging a message is $\varepsilon_1 = Pr[E_f]Pr[E_a | E_f]$, and the calculation process is as follows:

Claim 1: $Pr[E_f] = Pr[\neg bad] \geq 1 - \frac{(q_{H_0} + q_{H_2} + q_E)q_S}{q}$

Proof: If the values generated during the signature simulation, namely $\{m_i\}$, $\{m_i, A_i, VPK_i, \sigma, L, T_i\}$, and $\{VPK_i\}$, happen to occur in previous $Hash_0$ -Oracle, $Hash_2$ -Oracle, and $Extract Verification Key$ -Oracle queries, respectively, we set the flag $bad \leftarrow true$. Since the maximum number of requests is $q_{H_0} + q_{H_2} + q_E$, the probability of the aforementioned events

occurring for *Sign-Oracle* is at most $\frac{q_{H_0+q_{H_2}+q_E}}{q}$. Therefore, the maximum probability of the *Sign-Oracle* query event occurring is $\frac{(q_{H_0+q_{H_2}+q_E)q_S}}{q}$.

Claim 2: $Pr[E_a | E_f] \geq \varepsilon'$

Proof: If the challenger $\mathcal{C}$ is not terminated due to the *Sign-Oracle* query of the adversary $\mathcal{A}$, then $\mathcal{C}$'s responses to all queries are valid. Therefore, it is assumed that the probability of $\mathcal{A}$ generating a valid forgery is at least ε' .

Therefore, the non-negligible probability of the adversary $\mathcal{A}$ generating a valid forgery is at least $\varepsilon_1 = \varepsilon' - \frac{(q_{H_0+q_{H_2}+q_E)q_S}}{q}$. Subsequently, the challenger $\mathcal{C}$ utilizes *Hash₂-Oracle* request to produce two different outputs and runs a Forking algorithm [35] to obtain two valid signatures sig_i^1 and sig_i^2 , satisfying the equations $A_i = -sig_i^1 P + VPK_i \cdot l_2$ and $A_i = -sig_i^2 P + VPK_i \cdot l'_2$. Subtract them to get $(sig_i^2 - sig_i^1)P = (l_2 - l'_2)usk_i \cdot P$, and further successfully break ECDLP according to $usk_i = (l_2 - l'_2)^{-1}(sig_i^2 - sig_i^1) \bmod q$. According to $frk \geq acc \cdot (\frac{acc}{q} - \frac{1}{h})$ in the Forking Lemma, the probability of an effective solution of the output ECDLP can be obtained, as shown in formula (9). The running time for the challenger $\mathcal{C}$ in the above process is twice the time required by the adversary $\mathcal{A}$ for the attack, plus the sum of the time required for hash and signature queries. Therefore, we obtain the running time for $\mathcal{C}$ to output a valid solution for the ECDLP is $\tau \approx 2\tau' + 2t_h + (q_S + 1)(4t_{ecm} + 2t_{epa})$.

$$\varepsilon \geq \left(\varepsilon' - \frac{(q_{H_0} + q_{H_2} + q_E)q_S}{q} \right) \cdot \left(\frac{\varepsilon' - \frac{(q_{H_0+q_{H_2}+q_E)q_S}}{q}}{q_{H_0} + q_{H_2} + q_E} - \frac{1}{q} \right) \quad (9)$$

Thus, the proof is concluded.

B. Informal Theoretical Security Analysis

This subsection provides a detailed analysis of various security properties satisfied by our proposed scheme in a non-mathematical way.

1) *Message integrity and authenticity:* For the message reported by the vehicle, the vehicle needs to register the public key VPK_i before reporting the message, and a one-time anonymous signature sig_i (using the private key usk_i corresponding to VPK_i) can be generated after the successful registration in the blockchain, combined with the threshold signature σ to ensure the reported message reliability. According to

Theorem 1, if ECDLP is difficult to solve, attackers cannot obtain the private key usk_i , and under the protection of the weighting mechanism, they cannot acquire a sufficient number of secret shares s_i . Therefore, any Probabilistic Polynomial-Time (PPT) adversary cannot forge multiple signatures for messages or signatures for message reporting. Thus, the receiver Fog_j confirms the received message by checking the equations $sig_i P \stackrel{?}{=} A_i + VPK_i \cdot hs_i$ and $(l_x, l_y) = \sigma^{-1} \cdot H_0(m_i) \times P + \sigma^{-1} \cdot l_x \times pk_{f_j}$, while it can confirm the validity of the VPK_i by querying the blockchain. The former

equation ensures that the sender is legitimate, while the latter equation verifies the reliability of the reported message.

2) *Identity privacy-preserving:* During the message transmission process, vehicles only use the communication public key VPK_i registered in the blockchain and the corresponding private key usk_i to sign and encrypt messages, without revealing their real identities. Except for the public key owner and TA, no other entities can get the vehicle real identity based on publicly available message. Additionally, vehicles employ one-time anonymous public keys for communication to avoid being tracked by other vehicles or attackers.

3) *Traceability:* TA can obtain the blockchain address $address_i$ that registered the vehicle's public key VPK_i , and then retrieve the real identity based on the data stored in the database. Thus, TA can trace any illegal message sources in our scheme.

4) *Effective Revocation of Identity:* The malicious vehicle should be effectively revoked after its identity is traced. Our scheme is supplemented by the definition in literature [36], that is, the vehicle whose identity is revoked should not be able to continue to transfer the generated pseudonym or public and private keys to other legal or illegal vehicles, and enable the recipient of the pseudonym or public key to continue to generate valid signatures to communicate with other entities in VANETs. In our scheme, blockchain and the cooperation of the TA are used to revoke illegal identities. TA sends a request to revoke the corresponding blockchain address $address_i$ and all the communication public keys VPK_i^M registered under that address. Even if an illegal vehicle continues to generate valid signatures or transfers them to other users, the receiver cannot find a valid public key in the blockchain, making the signed message fail to be verified. Therefore, our scheme can effectively revoke identity.

5) *Resistance various attacks:* Our proposed scheme should be able to resist attacks such as vehicle collusion attacks, rogue key attacks and common types of attacks like impersonation attacks, modification attacks, replay attacks, and man-in-the-middle attacks, to ensure the scheme security.

• Resistance vehicle collusion attack

Suppose $t - 1$ legitimate but dishonest vehicles collude to attack an honest vehicle in an attempt to forge a valid signature (with a total weight reaching the threshold of t signatures), we will discuss the following two case:

Case 1: There are $t - 1$ attackers forge t invalid individual signatures, but the aggregation of t signatures can be verified. In our scheme, through the use of weighted threshold signature technology, each signer with a different weight utilizes the pre-distributed secret share s_μ to generate an aggregated signature by individually signing the message. It is impossible for any PPT attacker to forge a single valid signature without the key, further enabling the aggregated signature to pass verification, as the aggregated signature cannot pass the check of the verification key pk_{f_j} .

Case 2: There are $t - 1$ attackers collude to sign false messages reported by dishonest vehicles, thus passing the verification of Fog_j . According to the three-factor reputation evaluation design of our scheme, vehicles with higher weight values also

have higher reputation values. By controlling the reputation value through a "rapid decline and slow growth" approach, vehicles with higher reputation values need more positive behaviors to obtain rewards, but there are greater penalties for improper behaviors. Additionally, integrating improper behavior reporting mechanism into reputation evaluation can significantly reduce the possibility of attackers colluding to sign false messages.

- Resistance rogue key attack

This attack indicates that attackers can forge valid signatures by collecting legitimate public keys of other vehicles. However, in message reporting process, vehicles in our scheme utilize the threshold signature σ calculated based on the secret shares assigned to legitimate vehicles within the vehicle group before message reporting and the signature sig_i calculated using the communication private key. The verification process only involves the verification key pk_{f_j} and the communication public key VPK_i of the reporting vehicle, and does not rely on the aggregation of multiple public keys to verify the legitimacy of identity and messages. Therefore, attackers cannot launch rogue key attacks.

- Resistance other attacks

a) Resistance replay attack: In our scheme, all transmitted messages contain a timestamp, and the message receiver will check the freshness of the message by verifying whether the equation $T_{cur} - T_i \leq \Delta T$ holds before authenticating the identity. If the message has expired, the receiver will discard the message directly.

b) Resistance impersonation attack: Assuming that an attacker wants to impersonate a legitimate vehicle reporting an message, the attacker should construct the signatures σ and sig_i , and ensure that the equation $sig_i P \stackrel{?}{=} A_i + VPK_i \cdot hs_i$ holds. According to Theorem 1, the attacker cannot obtain the private key usk_i , and under the protection of the weighting mechanism, they cannot acquire a sufficient number of secret shares s_i to construct a valid signature. Therefore, our proposed scheme can resist impersonation attacks.

c) Resistance modification attack: According to Theorem 1, if an attacker modifies any parameters in $M_5 = \{m_i, A_i, VPK_i, \sigma, L, T_i, sig_i\}$, the receiver can determine that M_5 have been modified by judging that the equation $sig_i P \stackrel{?}{=} A_i + VPK_i \cdot hs_i$ does not hold, thus resisting modification attacks.

d) Resistance man-in-the-middle attack: In our scheme, messages sent to fog node Fog_j are signed using a one-time signature algorithm (the security of the signature is proven in Theorem 1), and transaction messages sent to the blockchain are signed using the ECDSA algorithm. Therefore, our proposed scheme can resist man-in-the-middle attacks.

6) Resistance side-channel and physical attacks: Our scheme design does not have any long-term keys, and the stored keys are updated periodically, specifically: 1. The threshold signature key s_i and two intermediate values (σ_i^0, σ_i^1) of the ECDSA algorithm stored at the vehicle side will be updated with the changes of the vehicle group. 2. The key usk_i used for communication at the vehicle side is one-time key. 3. The

TABLE III
COMPARISON OF FUNCTIONALITY FEATURES

	Li et al. [10]	Chen et al. [11]	Baza et al. [14]	Jiang et al. [37]	Srivastava et al. [38]	Our proposed
SF-1	✓	×	×	✓	✓	✓
SF-2	✓	✓	×	✓	✓	✓
SF-3	×	✓	×	×	×	✓
SF-4	✓	✓	✓	×	×	✓
SF-5	✓	✓	✓	✓	✓	✓
SF-6	✓	×	✓	×	×	✓
SF-7	×	✓	✓	✓	✓	✓
SF-8	✓	✓	✓	✓	✓	✓
SF-9	✓	✓	✓	✓	✓	✓

communication private key stored at the fog node side fsk_j is also updated periodically and the corresponding public key is published in the blockchain. By limiting the key lifecycle, most of these attacks can be rendered ineffective or have very limited impact. In addition, for the vehicle side, the signatures of the reported messages are designed in a threshold manner, so that even if a small number of vehicles have their threshold signing key s_i or the intermediate value of the signature, (σ_i^0, σ_i^1) , compromised by an attack before updating, the attacker is unable to generate a valid aggregated signature for the message (since the signature numbers cannot reach the threshold). Therefore, our scheme achieves resistance to side-channel and physical attacks through periodic key updates and threshold design.

C. Comparison of Security and Functionality

Table III presents a comparison of functional characteristics between existing schemes [10], [11], [14] and our proposed scheme. The symbol "✓" indicates that the scheme is able to resist attacks or possess specific functions, while the symbol "×" indicates that it is unable to resist attacks or lacks such functions. According to the Table III, our scheme provides more functions and better security features compared with the existing schemes [10], [11], [14]. Among them, we use SF-1, SF-2, SF-3, SF-4, SF-5, SF-6, SF-7, SF-8, SF-9 represents identity privacy-preserving, traceability, effective revocation of identity, resistance vehicle collusion attack, resistance rogue key attack, resistance replay attack, resistance impersonation attack, resistance modification attack and resistance man-in-the-middle attack, respectively.

VI. PERFORMANCE ANALYSIS

In this section, we first provide a brief description of the workflows of five baseline methods, followed by an analysis of the performance of our proposed scheme. Next, we compare the signature generation flexibility of our scheme with those proposed by [10], [11], [14], [37] and [38]. Finally, we analyze and compare the computational cost, communication cost, and network performance of those schemes.

For schemes [10] and [11] based on bilinear pairing, in order to achieve the 128-bit security level, we construct a bilinear pairing $\bar{e} = \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_T$, where $\bar{q}$ -order $\mathbb{G}_1$ is the additive group generated by point $\bar{P}$ on the supersingular elliptic curve $\bar{E} : y^2 = x^3 + x \mod \bar{p}$, where $\bar{p}$ and $\bar{q}$ are 512-bit primes. For

the scheme [14], [37], [38] and our scheme, we construct an additive group $\mathbb{G}$ generated by a point P of q -order on a non-singular elliptic curve $E : y^2 = x^3 + ax + b \bmod p$, where p and q are 256-bit primes, and set $p_0 = q$.

A. Baseline Methods Description

To demonstrate the advantages of our proposed scheme, we compare it with existing schemes [10], [11], [14], [37] and [38]. The descriptions of all comparison schemes are as follows:

- In Li et al.'s scheme [10], when the proxy vehicle V_i broadcasts a traffic message, surrounding vehicles V_l assist in signing the message using a threshold signature technique and respond. Once V_i collects more than t signatures, it generates a complete announcement message and sends it to the RSU. The RSU verifies the message using the blockchain, thus completing one round of message reporting.
- In Chen et al.'s scheme [11], when a vehicle V_i encounters a traffic accident, it first reports the emergency message to the RSU. Then, the RSU broadcasts the message to inquire which vehicles are willing to participate in the authentication process. The vehicles V_l that agree to assist interact with each other to generate a multi-signature and send it to the RSU. Once the RSU successfully verifies the signature, it reports the accident message to the transportation authority.
- In Baza et al.'s scheme [14], when a vehicle V_i needs to report an event, it requests location proofs from all RSUs within its communication range. Multiple consecutive RSUs generate proofs of the vehicle's anonymous locations using a threshold signature technique to construct a trajectory. After receiving the location proofs from the RSUs, V_i sends the latest trajectory along with the reported event to the event manager.
- In Jiang et al.'s scheme [37], multiple users sign a transaction using a multi-signature technique, where the sizes of the aggregated public key and aggregated signature are independent of the number of signers. Upon receiving the aggregated signature, the verifier performs verification, and if it is successful, the transaction is accepted.
- In Srivastava et al.'s scheme [38], a cluster head V_i is selected from the vehicle cluster. When V_i needs to report a message, it first generates a multi-signature with the help of other vehicles V_l within the cluster and submits the message along with the signature to the RSU. The RSU forwards the message to the cloud server after successful verification.

B. Our Scheme Performance Analysis

To analyze the computational cost of our proposed scheme on the vehicle side and the fog node, we implemented the entire cryptographic protocol using the MIRACL library and tested it on a personal computer (serving as the fog node, configured with an i7 11700X CPU @2.50 GHz, 16 GB RAM, and Ubuntu20.04 operating system) and on a Raspberry Pi 3B (serving as the vehicle side, configured with a Quad-core ARM Cortex-A53 CPU @ 1.2 GHz and 1 GB LPDDR2 SDRAM) respectively. The average

computational cost was obtained by testing 5000 times [39]. It is worth noting that we are testing the computational cost of pre-computing, single vehicle side and fog node side completion of a round of real-time message reporting and validation processes for all vehicle interactions in vehicle group. The complete code can be found at <https://github.com/beautylr/sugar>.

1) *Threshold signature cost analysis*: Since the threshold signature technology adopted in schemes [10], [11], [14], [37] and [38] does not lead to changes in the number and cost of threshold signatures, while our scheme's weighted threshold signature technology sees dynamic changes in the number and cost of threshold signatures with each signer's weight variation. Taking the threshold signature size of $b = 160$ bits (20 bytes) in scheme [10] as a reference, as shown in Fig. 6, we tested the threshold signature cost $|\sigma| = \sum_{x=1}^t w_x$ (the size of σ) of our scheme under dynamic changes in the weight values w_x of each participant as the number of iterations increases, when the total weight value is set to $W_n = 800, 1600, 2400, 3200$ (where n is the total number of vehicles, taking values of 5, 10, 15, 20, and $W_n = n * b = \sum_{x=1}^n w_x$), and a weight threshold $W_t = W_n/2$ is chosen, where, the signature size of the x -th signer is w_x bits, and t represents the number of vehicles participating in the threshold signature.

As can be seen from Fig. 6, when $n = 15$ (W_n approaching 2400 bits, $W_t=1200$ bits), the range of the threshold signature cost $|\sigma|$ of our scheme is between [153,164] bytes (i.e., [1224,1312] bits) in 100 iterations, and its floating range is within 11 bytes. The total communication cost is calculated by taking the average value 159 bytes of the threshold signature size, and the t takes the average value of 8. Similarly, when $n = 5, 10, 20$ (W_n approaches 800, 1600, 3200 bits, $W_t=400, 800, 1600$ bits), the floating values are within the acceptable range. The total communication cost is calculated by taking its average value of 60, 109, 210 bytes and t by taking its average value of 3, 5, 8, 10.5, respectively. The stability of our scheme is observed by testing the fluctuation of the threshold signature cost under different total numbers of vehicles, and it can be seen that the fluctuation of the cost is basically stabilized in a range under the test of 100 iterations.

2) *Computation cost analysis*: As shown in Fig. 7(a), the overall computational cost at the vehicle side and the fog node side is tested for different weight thresholds as the number of vehicles increases (total vehicle weight value W_n) in the vehicle group. The weight thresholds W_t are set as $1/4W_n, 1/3W_n, 1/2W_n, 2/3W_n, 3/4W_n$. As can be seen from the figure, the overall computational cost of the cryptographic algorithm at both ends increases approximately linearly with the increase in the number of vehicles, but when the number of vehicles is increased to 100, the cost is maximal at 266.5697ms and is controlled within 270 ms.

Similarly, as in Fig. 7(b), to test the overall computational cost at the vehicle side and fog node side with increasing threshold value for different number of vehicles, the growth of the threshold is set to increase linearly, i.e., evenly divided into five parts between the range of $[1/4W_n, 3/4W_n]$. As can be seen from the figure, the computational cost grows very little as the weight threshold increases, and the overall computational

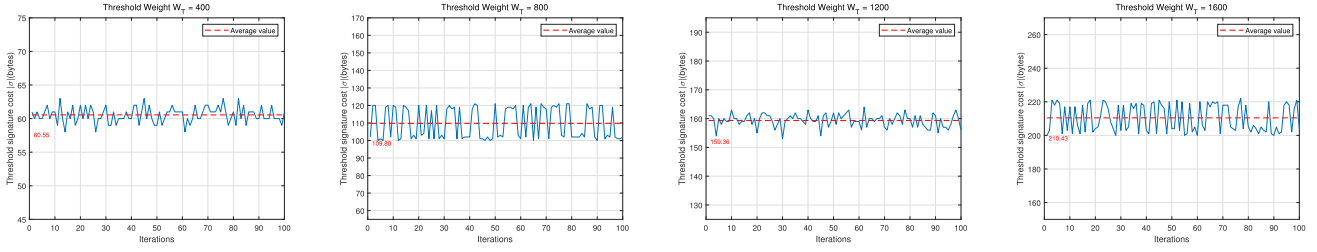
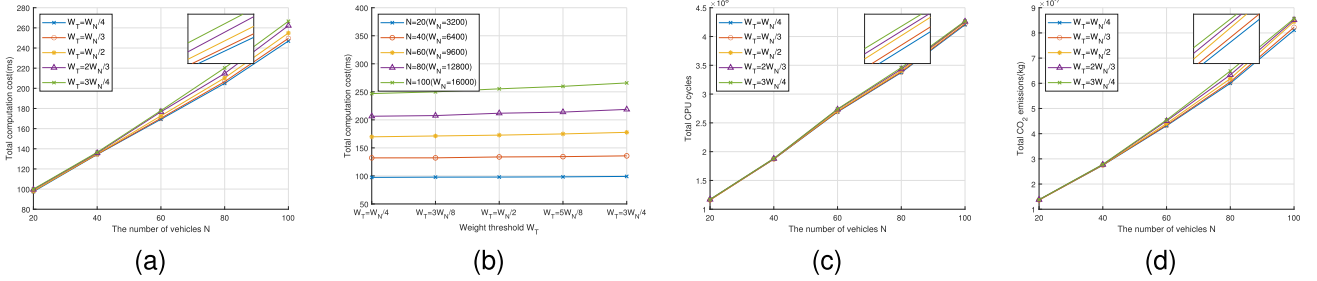


Fig. 6. The threshold signature cost of our scheme fluctuates.

Fig. 7. Performance analysis of our scheme (on vehicle-side and fog-side). (a) Total computation cost (varies with number of vehicles). (b) Total computation cost (varies with threshold). (c) Total CPU cycles. (d) Total CO_2 emissions.

cost increases significantly for the same weight threshold with different number of vehicles settings.

3) *Energy consumption analysis*: To quantitatively analyze the energy consumption of our scheme, we tested the number of CPU cycles and CO_2 emissions for the cryptographic protocols running at the vehicle side and the fog node side, respectively. Equation (10) is utilized to estimate the CO_2 emission of the device side in the specified time, where $Elec$ denotes electricity consumption, Emi_{per} denotes CO_2 emissions per kWh of electricity. In particular, the CO_2 emission per kWh is taken as the average value of 0.0608kg/kWh for China's provinces, which means that each kWh of electricity consumed emits about 0.608kg of CO_2 .

$$Emi_{CO_2} = Elec(kWh) \times Emi_{per}(kg/kWh) \quad (10)$$

As shown in Fig. 7(c) and (d), the cpu cycles and CO_2 emissions at the vehicle side and fog node side are tested with different weight thresholds as the number of vehicles (total vehicle weight value W_n) increases in the vehicle group. The weight threshold settings are the same as in Fig. 7(b). As can be seen from the figure, the overall computational cost of the cryptographic protocols at the vehicle-side and fog node-side grows approximately linearly with the increase in the number of vehicles, and when the number of vehicles is increased to 100, the maximum CO_2 emission is 8.5667×10^{-7} kg.

C. Comparative Analysis of Signature Generation Flexibility

To demonstrate the flexibility of our scheme in the signature generation process, we conducted tests under different weight threshold selections with a fixed number of vehicles $n = 10$ and

TABLE IV
COMPARISON OF t AND t^*

W_t	t^*	t	W_t	t^*	t	W_t	t^*	t
310	2	2.11	630	4	4.18	950	6	6.1
320	2	2.42	640	4	4.38	960	6	6.38
330	3	2.8	650	5	4.7	970	7	6.69
340	3	2.96	660	5	4.93	980	7	6.9
470	3	3.21	790	5	5.16	1110	7	7.1
480	3	3.5	800	5	5.41	1120	7	7.26
490	4	3.78	810	6	5.63	1130	8	7.69
500	4	3.97	820	6	5.88	1140	8	7.86

total weight value $W_n = 1600$. After 100 iterations of testing, we compared the average number t of threshold signature vehicles in our scheme with the value t^* of threshold signature vehicles in schemes [10], [11], [14], [37] and [38]. As shown in Table IV, by flexibly adjusting the weight threshold, the average number t of signature vehicles required by our scheme is lower than the t^* value of other comparison schemes, which not only reduces the signature cost, but can effectively resist the attack of some malicious nodes with low reputation.

It is worth noting that when a vehicle enters within the range of a new fog node Fog_j , even if there are slight changes in members, the fog node can dynamically adjust the threshold on demand (for example, based on current traffic flow). Only when the changes in members within the vehicle group fail to satisfy the safety policies will the fog node regroup the members into a new group and redistribute the weight values. Moreover, since vehicle group formation and weight value distribution can be completed before the vehicle reports messages, the dynamic nature of this process will not affect the stability of the system's reported messages.

TABLE V
THE TOTAL COMMUNICATION PACKET SIZE

	Packet size for reporting messages (bytes)
Li et al. [10]	$340 + 244t^*$
Chen et al. [11]	$264 + 128t^* + 512(t^* - 1)$
Baza et al. [14]	$256t^* + 96k + 924$
Jiang et al. [37]	$192t^*$
Srivastava et al. [38]	$4y comment + (3u + 2v)yt^*$
Our proposed	$356 + q + 108 \cdot t$

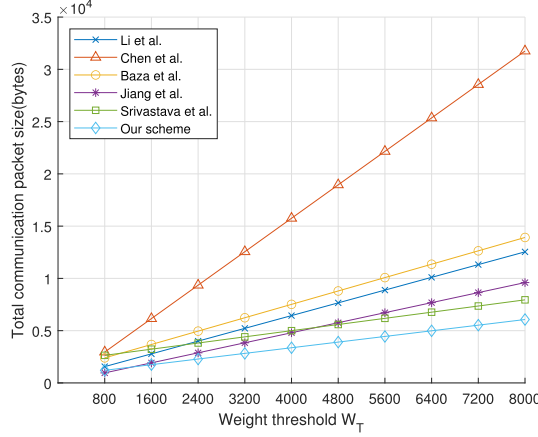


Fig. 8. Total communication packet size comparison.

D. Communication Cost Comparative Analysis

Since $\bar{q}$ and q are 64 bytes and 32 bytes respectively, the elements in group $\mathbb{G}_1$ and $\mathbb{G}$ are $64 \times 2 = 128$ bytes and $32 \times 2 = 64$ bytes respectively. Assuming that the output size of an ECDSA signature is 64 bytes, a hash function is 32 bytes, symmetric encryption/decryption and MAC function are both 20 bytes, the timestamp and the request are both 4 bytes, and the message is standardized as 20 bytes, we will only compare the message transmission length during the message reporting phase.

Table V shows the communication packet sizes of different schemes, where $|q|$ is the binary length of q . To visually display the comparison results, k represents the number of vehicles passing through the RSU, y denotes the number of rounds of the underlying recognition scheme, $|comment|$ denotes the size of the commitment scheme, u denotes the number of equations, and v denotes the number of variables, we set k, y, u, v are the minimum value 2, 2, 11, 13 and $|comment|$ take 256 bytes. Comparing the total communication cost of schemes [10], [11], [14], [37], [38] and our proposed scheme under different weight thresholds, as illustrated in Fig. 8. The total communication cost of our scheme is the average of 100 tests. Since the participating vehicles in schemes [10], [11], [14], [37] and [38] are equal, they can be regarded as having equal weights. Taking $b = 160$ bits (20 bytes) as a reference, the weight value for each party is set as 160 bits (20 bytes). When the weight threshold $W_t = 800, 1600, 2400, 3200, 4000, 4800, 5600, 6400, 7200, 8000$ (half of the total weight, $W_t = W_n/2$), $t^* = 5, 10, 15, 20, 25, 30, 35, 40, 45, 50$, $t = 5.5, 10.5, 15.5, 20.5, 25.5, 30.5, 35.5, 40.5, 45.5, 50.5$.

As can be seen from Fig. 8, the communication cost of Chen et al.'s scheme [11] increases rapidly with the increase in the threshold value, mainly due to the need to transmit the public keys of all participating signers during the message signature collection and reporting process. Compared with scheme [11], the cost of scheme [10], scheme [14], scheme [37] and scheme [38] is lower. However, our proposed scheme has the lowest cost among five schemes, mainly due to the use of weighted threshold technology and blockchain technology, which reduces the cost of designing vehicle identification and threshold signature.

E. Computational Cost Comparative Analysis

To analyze and compare the computational cost of our proposed scheme and other existing schemes at the vehicle side and fog node, we employed the MIRACL library to perform various cryptographic operations 5000 times on a personal computer and a Raspberry Pi 3B. The average execution times were obtained [39]. Table VI shows the execution time of related operations, where $Time_1$ represents the execution time at the vehicle side and $Time_2$ represents the execution time at the fog node. Note that we ignore the time to perform the Xor, small scalar multiplication operation due to their relatively short execution time compared to other operations.

We conduct a theoretical analysis of our scheme and several existing schemes from three aspects: message reporting, message verification, and batch message verification, as shown in Table VII. where b represents the number of messages stored in the block and the value is 3000, n indicates the number of messages that RSU verifies in batches.

To facilitate observation, based on the test setup in terms of communication cost, the computational costs in signing, verification, and total computational costs of the schemes [10], [11], [14], [37], [38], and our proposed scheme are compared under different weight thresholds, as shown in Fig. 9(a), (b), and (c) respectively. As can be seen from Fig. 9(a), the computational cost of vehicle-side message reporting will increase linearly with the increase of the weight threshold, and our scheme has the slowest growth rate and the lowest cost. According to Fig. 9(b), the message verification computational costs of all schemes remain constant with the increase in weight threshold. Our proposed scheme has a slightly higher cost than the schemes [11] and [37], but lower than the schemes [10], [14] and [38]. As can be seen from Fig. 9(c), our proposed scheme has the lowest total computational cost, outperforming other existing schemes.

Fig. 9(d) shows the execution time for batch verification of different numbers of reported messages. It can be observed that compared with the schemes [10] and [14], our scheme achieves higher performance in batch verification. Although our scheme has a higher cost for verifying a single message compared with the scheme [11], the cost for batch message verification is lower, and the advantage gradually increases with the number of messages.

TABLE VI
THE EXECUTION TIME OF CRYPTOGRAPHIC OPERATIONS

Symbol	Description	Format	$Time_1$ (ms)	$Time_2$ (ms)
T_{bp}	Bilinear pairing operation	$e(g_1, g_2)$, where $g_1, g_2 \in G_1$	54.801	3.795
T_{bsm}	Scale multiplication operation on G_1	$g^x, g \in G_1, x \in Z_p^*$	9.911	0.515
T_{bpa}	Point addition operation on G_1	$S + T$, where $S, T \in G_1$	0.022	0.005
T_{ecm}	Scale multiplication operation on G	$x \cdot P$, where $P \in G$ and $x \in Z_p^*$	8.4572	0.4148
T_{epa}	Point addition operation on G	$S + T$, where $S, T \in G$	0.0479	0.0023
T_{mac}	MAC signature operation	$H(rk \parallel \cdot)$, where $rk \in Z_q^*$	0.1044	0.0064
T_{mtp}	Map-to-point hash operation	$H : \{0, 1\}^* \rightarrow G_1$	1.302	0.042
T_{aes}	Symmetric encryption/decryption operation(AES-CBC)	$E(k, \cdot) / D(k, \cdot)$	0.0574	0.0021
T_h	One-way hash function operation	$H(\cdot)$	0.034	0.0016
T_{com}	Commitment scheme operation	$g^m \cdot h^r, g, h \in G, m, r \in Z_p^*$	16.914	0.8296

TABLE VII
COMPARISON OF COMPUTATIONAL COST

Scheme	Message report (vehicle/user)	Message verification (Fog node/RSU)	Message batch verification
[10]	$2t^*T_{bp} + (1+5t^*)T_{bsm} + (3t^*-1)T_{bpa} + (1+2t^*)T_h + t^*T_{mtp} + (t^*+1)T_{mac}$	$2T_{bp} + (\lceil \log_2 b \rceil + 1)T_h + T_{mtp}$	$2nT_{bp} + (\lceil \log_2 b \rceil + 1)nT_h + nT_{mtp}$
[11]	$2t^*T_{bsm} + t^*T_{bpa} + (t^*+1)T_h$	$2T_{bsm} + T_{bpa}$	$(n+1)T_{bsm} + (3n+1)T_{bpa}$
[14]	$2t^*T_{bp} + (2t^*+2)T_{bsm} + (t^*-1)T_{bpa} + (t^*+2)T_{mtp} + T_h$	$4T_{bp} + T_{mtp}$	$4nT_{bp} + nT_{mtp}$
[37]	$3t^*T_h + (2(t^*)^2 + t)T_{ecm}$	$T_h + 2T_{ecm}$	/
[38]	$5t^*T_{com} + T_h + t^*T_{epa}$	$10T_{com}$	/
Ours	$(2t+2)T_{mac} + (2t+1)T_{ecm} + (t+2)T_h + 2tT_{aes}$	$4T_{ecm} + 2T_{epa} + 2T_h$	$(n+3)T_{ecm} + (3n-1)T_{epa} + 2nT_h$

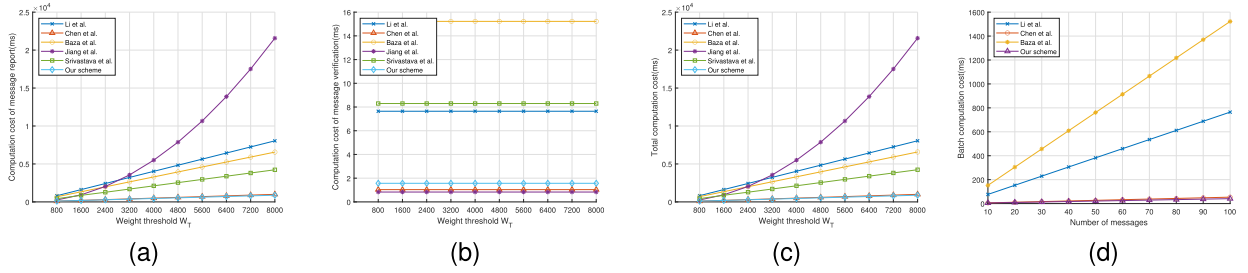


Fig. 9. Computational cost analysis. (a) Message report computation cost comparison. (b) Message verification computation cost comparison. (c) Comparison of total computing cost. (d) Message batch verification computation cost comparison.

F. Network Simulation Analysis

In this section, the network performance is tested using the Veins 4.7 framework, which serves as an intermediate layer between the network simulator Omnet++ and the traffic flow simulator Sumo. A comparative analysis is conducted on the network delay of our proposed scheme with the schemes [10], [11], [14], [37] and [38]. The relevant parameters are presented in Table VIII.

The analysis of the average transmission delay for vehicles to successfully report messages is defined as in (11), where n represents vehicles number, while n_i represents the number of messages reported by vehicle V_i . t_s^j , t_r^j and t_{rep} represent the time of sending, receiving the message report and completing the computation of operations related to the message report, respectively. As shown in Fig. 10(a), the transmission delay under different vehicle speeds is tested with the increase of the number of vehicles. It can be seen that as the density of vehicles increases, the transmission delay at different speeds grows, but with the same number of vehicles, the transmission delay fluctuates less when setting different speeds, which shows that the transmission delay is less affected by the speed of

TABLE VIII
SIMULATION PARAMETER

Parameter	Value
Simulation area	2500×2500 (m^2)
Data transmission rate	6 Mbps
Transmission power	50 mW
Sensitivity	-89 dBm
Path loss model	Two-Ray interference
Obstacle shadowing model	Simple obstacle shadowing
Max interference distance	2600 m
Thermal noise	-110 dBm
Used channel	CCH
Max speed	40 m/s
Acceleration speed of vehicle	10 m/s^2
Simulation time	200 s

vehicles. From Fig. 10(b), the vehicle speed is set at 20m/s to test the transmission delay of our scheme under different weight threshold settings. Based on the analysis in Section IV-A, the number of signature vehicles reaching the weight threshold W_t can be obtained. It can be observed that, with the same vehicle density n , the larger the weight threshold W_t is set (e.g., $\frac{1}{4}W_n$, $\frac{1}{3}W_n$, $\frac{1}{2}W_n$, $\frac{2}{3}W_n$, $\frac{3}{4}W_n$), the longer the time delay

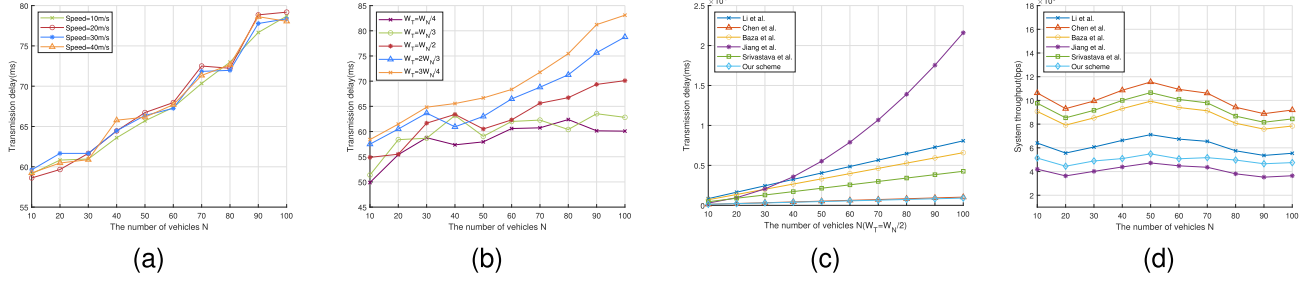


Fig. 10. Network performance analysis. (a) Transmission delay under different speed. (b) Transmission delay under different thresholds. (c) Comparison of transmission latency. (d) Comparison of system throughput.

required for vehicles to generate message reports.

$$A_{TD} = \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{n_i} \sum_{j=1}^{n_i} (t_r^j - t_s^j + t_{rep}) \right) \quad (11)$$

As shown in Fig. 10(c), as the total number of vehicles n increases from 10 to 100 (with the weight threshold set at half of the total weight, i.e., $W_t = \frac{W_n}{2}$), we compare the average transmission delay of our scheme with other schemes, where the calculation time for sending and receiving messages is included. The simulation results show that, compared with the other five schemes, the average transmission delay of our proposed scheme is the lowest.

The throughput of the system during the vehicle report message, defined as shown in (12), where Bit_n denotes the total number of bits of the message received during the simulation time and Sim_t denotes the system simulation time. As shown in Fig. 10(d), as the total number of vehicles n increases from 10 to 100, we compare the system throughput of our scheme with other schemes. From the simulation results, it can be seen that the system throughput of our scheme is higher than that of Jiang et al. [37] and lower than that of the other four schemes, which is basically in line with the fact that the communication cost and transmission delay are lower, and so is the system throughput.

$$A_{THP} = Bit_n / Sim_t \quad (12)$$

G. Blockchain Configuration and Consensus Optimization Discussion

1) *Blockchain Configuration*: To implement the blockchain-based prototype, our scheme primarily adopts the public Ethereum test network Goerli to deploy smart contracts. Goerli employs the Clique Proof-of-Authority (PoA) consensus mechanism, in which designated validators take turns to produce blocks without relying on mining. A new block is generated approximately every 15 seconds. The network's validator nodes are operated by multiple public infrastructure providers, and synchronization of blocks, state, and transaction data is conducted using the devp2p protocol. All gas computations, contract executions, and state transitions on Goerli are fully consistent with those of the Ethereum mainnet, making it a reliable environment for deployment testing, cost estimation, and debugging. We

connect Remix to Goerli via the MetaMask Chrome extension in Google Chrome and deploy the compiled smart contracts using Remix. This setup enables the measurement of the on-chain cost incurred by various VANET entities when performing operations such as registration, query, and revocation via smart contracts.

2) *Consensus Optimization Discussion*: To further address the limitations of blockchain throughput and latency in VANET applications, future improvements can be explored by optimizing the underlying consensus mechanism. For instance, from a game-theoretic perspective, a delegated-agent model involving vehicles and RSUs can be designed to reach a Nash equilibrium, allowing dynamic adjustment of the difficulty level in Proof of Work (PoW) consensus. Additionally, considering the profits obtained from block generation and stake investments, incentive allocation in Proof of Stake (PoS) consensus can be optimized using game theory to improve fairness and efficiency. From a cryptographic perspective, threshold signature techniques may be leveraged to reduce the communication cost of committee-based Byzantine Fault Tolerance (BFT) protocols, offering another promising direction for consensus optimization.

VII. CONCLUSION

To address the issues of flexibility, security and efficiency in message reporting of VANETs, a blockchain-assisted message reporting scheme with weighted threshold signatures for VANETs is proposed. Our proposed scheme designs a three-factor reputation evaluation method to assign vehicle weight values combined with a misbehavior reporting mechanism, to improve the credibility of reported messages. Threshold signature messages are flexibly generated and quickly reported using the weighted threshold signature technology, which makes this design particularly suitable for scenes with low vehicle densities. In addition, our proposed scheme utilizes a blockchain for vehicle identity management, in which vehicles register one-time anonymous public keys with the blockchain on demand for communication, thereby protecting identity privacy while resisting various attacks. In case of malicious behavior, vehicle identity can be effectively revoked to enhance system security by revoking the anonymous public key. Security analysis showed that our proposed scheme satisfies the security and privacy

requirements for message reporting in VANETs. In performance analysis, our scheme outperforms the most advanced message reporting schemes for VANETs in terms of transmission, computational performance, and practicability. In the future, we plan to optimize the blockchain's consensus mechanism and integrate a high-performance filter to design a more efficient and automated distributed public information management method, thereby enabling a more secure and reliable message reporting scheme for the VANETs.

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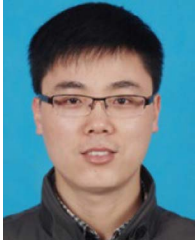
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Ru Li is currently working toward the PhD degree with the School of Computer Science and Technology, Anhui University, Hefei, China. Her research interests include vehicular ad hoc networks and applied cryptography.



Jie Cui (Senior Member, IEEE) was born in Henan Province, China, in 1980. He received the PhD degree from the University of Science and Technology of China in 2012. He is currently a professor and PhD supervisor with the School of Computer Science and Technology, Anhui University. He has more than 150 scientific publications in reputable journals such as *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Information Forensics and Security*, *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Mobile Computing*, *IEEE Transactions on Parallel and Distributed Systems*, *IEEE Transactions on Computers*, *IEEE Transactions on Vehicular Technology*, *IEEE Transactions on Intelligent Transportation Systems*, *IEEE Transactions on Network and Service Management*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Industrial Electronics*, *IEEE Transactions on Cloud Computing*, *IEEE Transactions on Multimedia*, academic books, and international conferences. His current research interests include applied cryptography, IoT security, vehicular ad hoc network, cloud computing security, and software-defined networking.



Jing Zhang was born in Henan Province, China, in 1990. She received the MA Eng and PhD degrees in computer science from Anhui University in 2021. She is currently an associate professor with the School of Computer Science and Technology, Anhui University. She has nearly 20 scientific publications in reputable journals such as *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Information Forensics and Security*, *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Vehicular Technology*, *IEEE Transactions on Intelligent Transportation Systems*, *Information Sciences*, *Science China Information Sciences*, *Vehicular Communications*, and international conferences. Her research interests include vehicular ad hoc network, IoT security, and applied cryptography.



Lu Wei (Member, IEEE) was born in Anhui Province, China, in 1993. He received the PhD degree in computer science from Anhui University in 2022. He is currently a lecturer with the School of Computer Science and Technology, Anhui University. He has more than 10 scientific publications in reputable journals such as *IEEE Transactions on Information Forensics and Security*, *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Mobile Computing*, and *IEEE Transactions on Intelligent Transportation Systems*. His research interests include vehicular ad hoc networks and applied cryptography.



Hong Zhong was born in Anhui Province, China, in 1965. She received the PhD degree in computer science from the University of Science and Technology of China in 2005. She is currently a professor and PhD supervisor with the School of Computer Science and Technology, Anhui University. She has more than 200 scientific publications in reputable journals such as *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Parallel and Distributed Systems*, *IEEE Transactions on Mobile Computing*, *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Intelligent Transportation Systems*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Vehicular Technology*, *IEEE Transactions on Network and Service Management*, *IEEE Transactions on Cloud Computing*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Industrial Electronics*, *IEEE Transactions on Big Data*, academic books, and international conferences. Her research interests include applied cryptography, IoT security, vehicular ad hoc network, cloud computing security, and software-defined networking.



Debiao He (Member, IEEE) received the PhD degree in applied mathematics from the School of Mathematics and Statistics, Wuhan University, Wuhan, China, in 2009. He is currently a professor with the School of Cyber Science and Engineering, Wuhan University. He has authored or coauthored more than 100 research papers in refereed international journals and conferences, such as *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Information Security and Forensic*, and *Usenix Security Symposium*. His main research interests include cryptography and information security, in particular, cryptographic protocols. He is the recipient of the 2018 IEEE Sysems Journal Best Paper Award and the 2019 IET Information Security Best Paper Award. His work has been cited more than 10000 times at Google Scholar. He is in the Editorial Board of several international journals, such as *Journal of Information Security and Applications*, *Frontiers of Computer Science*, and *Human-centric Computing & Information Sciences*.